



Introduction

A rectangular array of mn numbers (which may be **real or complex**) in the form of 'm' horizontal lines (called **rows**) and 'n' vertical lines (called **columns**), is called a matrix of order m by n, written as $m \times n$ matrix.

Such an array is enclosed by [] or (). An $m \times n$ matrix is usually written as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

In compact form, the above matrix is represented by $A = [a_{ij}]_{m \times n}$. The number $a_{11}, a_{12}, \dots$ etc are known as the elements of the matrix A, a_{ij} belongs to the i^{th} row and j^{th} column and is called the **(i, j)th** element of the matrix $A = [a_{ij}]$.

e.g., $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & 9 \end{bmatrix}$ is a matrix having 2 rows and 3 columns. Its order is 2×3 and it has 6 elements:

$a_{11} = 1, a_{12} = 2, a_{13} = 3, a_{21} = 0, a_{22} = -1, a_{23} = 9$.

Special Type of Matrices

(a) Row Matrix (Row vector)

$A = [a_{11}, a_{12}, \dots, a_{1n}]$ i.e. row matrix has exactly one row.

(b) Column Matrix (Column Vector)

$A = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}$ i.e. column matrix has exactly one column.

(c) Zero or Null Matrix

($A = \mathbf{O}_{m \times n}$) An $m \times n$ matrix whose all entries are zero.

$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ is a 3×2 null matrix and $B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is 3×3 null matrix

(d) Horizontal Matrix

A matrix of order $m \times n$ is a horizontal matrix if $n > m$

e.g., $\begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & 5 & 1 & 1 \end{bmatrix}$



(e) Vertical Matrix

A matrix of order $m \times n$ is a vertical matrix if $m > n$ e.g. $\begin{bmatrix} 2 & 5 \\ 1 & 1 \\ 3 & 6 \\ 2 & 4 \end{bmatrix}$

(f) Square Matrix

If number of rows = number of columns $\Rightarrow$ matrix is a square matrix.

If number of rows = number of columns = n then, matrix is of the **order 'n'**.

Note: The pair of elements a_{ij} and a_{ji} are called **Conjugate Elements**.

Trace of Matrix

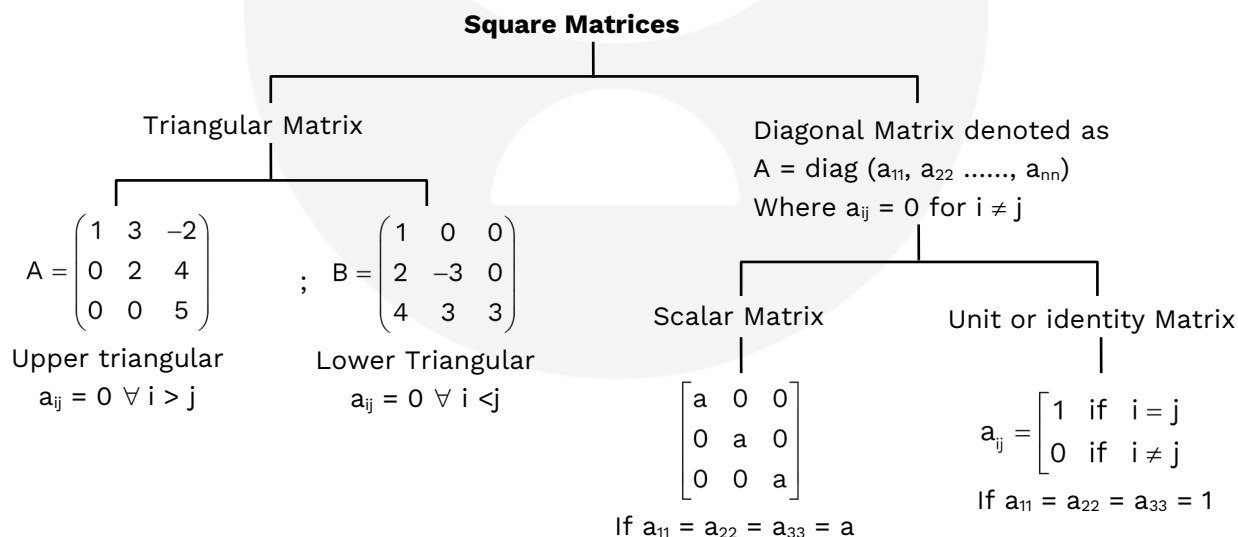
The sum of the elements of a **square matrix A** lying along the principal diagonal is called the trace of

A i.e. $(\text{tr}(A))$. Thus, if $A = [a_{ij}]_{n \times n}$, then $\text{tr}(A) = \sum_{i=1}^n a_{ii} = a_{11} + a_{22} + \dots + a_{nn}$

Properties of Trace of a Matrix

Let $A = [a_{ij}]_{n \times n}$ and $B = [b_{ij}]_{n \times n}$ and λ be a scalar then

- (i) $\text{tr}(\lambda A) = \lambda \text{tr}(A)$ (ii) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ (iii) $\text{tr}(AB) = \text{tr}(BA)$



- Note:** (i) Minimum number of zeros in triangular matrix of order $n = n(n-1)/2$.
 (ii) Minimum number of zero in a diagonal matrix of order $n = n(n-1)$.

Equality of Matrices

Let $A = [a_{ij}]$ and $B = [b_{ij}]$ are equal if

- (a) both have the same order.
 (b) $a_{ij} = b_{ij}$ for each pair of i and j .

**Example 1:**

Find the value of x , y , z and w which satisfy the matrix equation

$$\begin{bmatrix} x+3 & 2y+x \\ z-1 & 4w-8 \end{bmatrix} = \begin{bmatrix} -x-1 & 0 \\ 3 & 2w \end{bmatrix}$$

Solution:

As the given matrices are equal so their corresponding elements are equal.

$$x + 3 = -x - 1 \Rightarrow 2x = -4$$

$$\therefore x = -2 \quad \text{.....(i)}$$

$$2y + x = 0 \Rightarrow 2y - 2 = 0 \text{ [from (i)]}$$

$$\Rightarrow y = 1 \quad \text{.....(ii)}$$

$$z - 1 = 3 \Rightarrow z = 4 \quad \text{.....(iii)}$$

$$4w - 8 = 2w \Rightarrow 2w = 8$$

$$\therefore w = 4 \quad \text{.....(iv)}$$

Ans.**Concept Builders - 1**

- (i) Find 2×3 matrix $[a_{ij}]_{2 \times 3}$, where $a_{ij} = i + 2j$
- (ii) Find the minimum number of zeroes in a triangular matrix of order 4.
- (iii) Find minimum number of zeros in a diagonal matrix of order 6.
- (iv) If $\begin{bmatrix} 2x+y & 2 & x-2y \\ a-b & 2a+b & -3 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 4 \\ 4 & -1 & -3 \end{bmatrix}$, then find the value of x , y , a and b .

Algebra of Matrices**Addition**

$A + B = [a_{ij} + b_{ij}]$ where A and B are of the same order.

(a) Addition of Matrices Is Commutative

i.e. $A + B = B + A$

(b) Matrix Addition Is Associative

$(A + B) + C = A + (B + C)$

(c) Additive Inverse

If $A + B = \mathbf{O} = B + A$, then B is called **additive inverse** of A .

(d) Existence of Additive Identity

Let $A = [a_{ij}]$ be an $m \times n$ matrix and $\mathbf{O}$ be an $m \times n$ zero matrix, then $A + \mathbf{O} = \mathbf{O} + A = A$. In other words, $\mathbf{O}$ is the **additive identity** for matrix addition.

(e) Cancellation Laws

hold good in case of addition of matrices. If A, B, C are matrices of the same order, then

$A + B = A + C \Rightarrow B = C$ (**left cancellation law**)

and $B + A = C + A \Rightarrow B = C$ (**right-cancellation law**)

Note: The zero matrix plays the same role in matrix addition as the number zero does in addition of numbers.

**Example 2:**

If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = \mathbf{O}$ (zero matrix), then D matrix will be-

(A) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 6 & 5 \end{bmatrix}$

(B) $\begin{bmatrix} 0 & 2 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

(C) $\begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

(D) $\begin{bmatrix} 0 & -2 \\ -3 & -7 \\ -5 & -6 \end{bmatrix}$

Solution:

Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$$\therefore A + B - D = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{array}{lll} -a = 0 & \Rightarrow & a = 0, \\ 3 - c = 0 & \Rightarrow & c = 3, \\ 5 - e = 0 & \Rightarrow & e = 5, \end{array} \quad \begin{array}{lll} 1 - b = 0 & \Rightarrow & b = 1 \\ 7 - d = 0 & \Rightarrow & d = 7, \\ 6 - f = 0 & \Rightarrow & f = 6 \end{array}$$

$$\therefore D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$$

Ans. (C)**Concept Builders - 2**

(i) If $A = \begin{bmatrix} 2 & 3 & 9 \\ 8 & -2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & -7 & 2 \\ 6 & 4 & 8 \end{bmatrix}$, then find a matrix C such that $A - B + C = \mathbf{O}$ and also find the order of the matrix C.

(ii) If $A = \begin{bmatrix} 8 & 9 \\ 7/2 & 8 \\ 1 & -1 \end{bmatrix}$, then find the additive inverse of A and show that additive inverse of additive inverse will be the matrix itself.

Multiplication of a Matrix by a Scalar

$$\text{If } A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}; \quad kA = \begin{bmatrix} ka & kb & kc \\ kb & kc & ka \\ kc & ka & kb \end{bmatrix}$$

Properties of Scalar Multiplication

- (a) If A and B are two matrices of the same order and 'k' be a scalar then $k(A + B) = kA + kB$.
- (b) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 + k_2)A = k_1A + k_2A$.
- (c) If k_1 and k_2 are two scalars and 'A' is a matrix, then $(k_1 k_2)A = k_1(k_2A) = k_2(k_1A)$.



Multiplication of Matrices (Row by Column)

Let A be a matrix of order $m \times n$ and B be a matrix of order $p \times q$, then the matrix multiplication AB is possible if and only if $n = p$ and matrices are said to be **conformable** for multiplication.

In the product AB, A is called **pre-factor** and B is called **post factor**.

$\Rightarrow$ AB is possible if and only if **number of columns in pre-factor = number of rows in post-factor**.

Let $A_{m \times n} = [a_{ij}]$ and $B_{n \times p} = [b_{ij}]$, then order of AB is $m \times p$ and $(AB)_{ij} = \sum_{r=1}^n a_{ir}b_{rj}$

e.g., $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}_{2 \times 3}$ and $B = \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix}_{3 \times 4}$

Then order of AB is 2×4 .

$$(AB)_{11} = a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} = \sum_{r=1}^3 a_{1r}b_{r1}$$

$$(AB)_{23} = a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} = \sum_{r=1}^3 a_{2r}b_{r3}$$

$$\text{In general } (ab)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} = \sum_{r=1}^3 a_{ir}b_{rj}$$

Example 3:

If $\begin{bmatrix} 1 & x & 2 \end{bmatrix} \begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 2 \\ 0 & 3 & 2 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = \mathbf{O}$, then the value of x is:

- (A) -1 (B) 0 (C) 1 (D) 2

Solution:

The LHS of the equation

$$= \begin{bmatrix} 2 & 4x + 9 & 2x + 5 \end{bmatrix} \begin{bmatrix} x \\ 1 \\ -1 \end{bmatrix} = [2x + 4x + 9 - 2x - 5] = 4x + 4$$

$$\text{Thus } 4x + 4 = 0 \Rightarrow x = -1$$

Ans. (A)

Example 4:

If A, B are two matrices such that $A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$, $A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix}$, then find AB.

Solution:

$$\text{Given } A + B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \dots\dots(i) \quad \text{and} \quad A - B = \begin{bmatrix} 3 & 2 \\ -2 & 0 \end{bmatrix} \dots\dots(ii)$$

Adding (i) and (ii)

$$2A = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \Rightarrow A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$$

Subtracting (ii) from (i)

$$2B = \begin{bmatrix} -2 & 0 \\ 4 & 4 \end{bmatrix} \Rightarrow B = \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix} \quad \text{Now } AB = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 4 & 4 \end{bmatrix}$$

Ans.



Properties of Matrix Multiplication

(A) Matrix Multiplication is Not Commutative: I.E. $AB \neq BA$

Here both AB and BA exist and also, they are of the same type but $AB \neq BA$.

Example:

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; \text{ then } AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}; BA = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\Rightarrow AB \neq BA \quad (\text{in general})$$

(b) $AB = \mathbf{O} \Rightarrow A = \mathbf{O}$ or $B = \mathbf{O}$ (in general)

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \text{ then } AB = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Note: If A and B are two non-zero matrices such that $AB = \mathbf{O}$ then A and B are called the divisors of zero. If A and B are two matrices such that

- (i) $AB = BA$ then A and B are said to commute
- (ii) $AB = -BA$ then A and B are said to anticommute

(c) Matrix Multiplication Is Associative

If A , B and C are conformable for the product AB and BC , then $(AB)C = A(BC)$

(d) Distributivity

$$\left. \begin{array}{l} A(B+C) = AB + AC \\ (A+B)C = AC + BC \end{array} \right\} \text{ Provided } A, B \text{ and } C \text{ are conformable for respective products}$$

Example 5:

$$\text{Let } A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix}, B = \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} \text{ and } C = \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} \text{ be the matrices then, prove that}$$

in matrix multiplication cancellation law does not hold.

Solution:

We have to show that $AB = AC$; though B is not equal to C .

$$\text{We have } AB = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 1 & -2 & 1 & 2 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

$$\text{Now, } AC = \begin{bmatrix} 1 & -3 & 2 \\ 2 & 1 & -3 \\ 4 & -3 & -1 \end{bmatrix} \begin{bmatrix} 2 & 1 & -1 & -2 \\ 3 & -2 & -1 & -1 \\ 2 & -5 & -1 & 0 \end{bmatrix} = \begin{bmatrix} -3 & -3 & 0 & 1 \\ 1 & 15 & 0 & -5 \\ -3 & 15 & 0 & -5 \end{bmatrix}_{3 \times 4}$$

Here, $AB = AC$ though B is not equal to C . Thus, cancellation law does not hold in general.

Concept Builders - 3



- (i) If $A = \begin{bmatrix} 2 & 9 \\ -4 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 5 \\ 6 & 8 \end{bmatrix}$ and $C = \begin{bmatrix} 9 & -7 \\ -2 & 4 \end{bmatrix}$, then show that $A(B + C) = AB + AC$.
- (ii) If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -1 & 1 \end{bmatrix}$, then prove that $(A - B)^2 \neq A^2 - 2AB + B^2$.
- (iii) Find the value of x : $2 \begin{bmatrix} 3 & 1 & -2 \\ -1 & -3 & 4 \end{bmatrix} + x \begin{bmatrix} 1 & 0 & -1 \\ 3 & 4 & 5 \end{bmatrix} = \begin{bmatrix} 4 & 2 & -2 \\ -8 & -14 & -2 \end{bmatrix}$

Positive Integral Powers of a Square Matrix

For a square matrix A , $A^n = \underbrace{A.A.A \dots A}_{\text{upto } n \text{ times}}$, where $n \in \mathbb{N}$

- Note:** (i) $A^m \cdot A^n = A^{m+n}$
(ii) $(A^m)^n = A^{mn}$, where $m, n \in \mathbb{N}$
(iii) If A and B are square matrices of same order and $AB = BA$ then
 $(A + B)^n = {}^nC_0 A^n + {}^nC_1 A^{n-1} B + {}^nC_2 A^{n-2} B^2 + \dots + {}^nC_n B^n$
Note that for a unit matrix I of any order, $I^m = I$ for all $m \in \mathbb{N}$.

Concept Builders - 4



- (i) If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then prove that $A^n = \begin{bmatrix} \cos n\alpha & \sin n\alpha \\ -\sin n\alpha & \cos n\alpha \end{bmatrix}$ where n is positive integer

The Transpose of a Matrix (Changing Rows and Columns)

Let A be any matrix of order $m \times n$. Then A^T or $A' = [a_{ji}]$ for $1 \leq i \leq m$ and $1 \leq j \leq n$ of order $n \times m$

Properties of Transpose

If A^T and B^T denote the transpose of A and B ,

- (a) $(A + B)^T = A^T + B^T$; note that A and B have the same order.
(b) $(A B)^T = B^T A^T$ (Reversal law) A and B are conformable for matrix product AB

Note: In general: $(A_1 A_2 \dots A_n)^T = A_n^T \dots A_2^T A_1^T$ (reversal law for transpose)

- (c) $(A^T)^T = A$
(d) $(kA)^T = kA^T$, k is a scalar.

Example 6:

If A and B are matrices of order $m \times n$ and $n \times m$ respectively, then order of matrix $B^T(A^T)^T$ is—
(A) $m \times n$ (B) $m \times m$ (C) $n \times n$ (D) Not defined

Solution:

Order of B is $n \times m$ so order of B^T will be $m \times n$

Now $(A^T)^T = A$ and its order is $m \times n$. For the multiplication $B^T(A^T)^T$

Number of columns in prefactor $\neq$ Number of rows in post factor.

Hence this multiplication is not defined.

Ans. (D)



Orthogonal Matrix

A square matrix is said to be orthogonal matrix if $A A^T = I$

Note: (i) The determinant value of orthogonal matrix is either 1 or -1.

$$(ii) \quad \text{Let } A = \begin{bmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$$

$$AA^T = \begin{bmatrix} a_1^2 + a_2^2 + a_3^2 & a_1b_1 + a_2b_2 + a_3b_3 & a_1c_1 + a_2c_2 + a_3c_3 \\ b_1a_1 + b_2a_2 + b_3a_3 & b_1^2 + b_2^2 + b_3^2 & b_1c_1 + b_2c_2 + b_3c_3 \\ c_1a_1 + c_2a_2 + c_3a_3 & c_1b_1 + c_2b_2 + c_3b_3 & c_1^2 + c_2^2 + c_3^2 \end{bmatrix}$$

If $AA^T = I$, then

$$\sum_{i=1}^3 a_i^2 = \sum_{i=1}^3 b_i^2 = \sum_{i=1}^3 c_i^2 = 1 \text{ and } \sum_{i=1}^3 a_i b_i = \sum_{i=1}^3 b_i c_i = \sum_{i=1}^3 c_i a_i = 0$$

Example 7:

Determine the values of α, β, γ , when $\begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$ is orthogonal.

Solution:

$$\text{Let } A = \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix}$$

But given A is orthogonal.

$$\therefore AA^T = I$$

$$\Rightarrow \begin{bmatrix} 0 & 2\beta & \gamma \\ \alpha & \beta & -\gamma \\ \alpha & -\beta & \gamma \end{bmatrix} \begin{bmatrix} 0 & \alpha & \alpha \\ 2\beta & \beta & -\beta \\ \gamma & -\gamma & \gamma \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 4\beta^2 + \gamma^2 & 2\beta^2 - \gamma^2 & -2\beta^2 + \gamma^2 \\ 2\beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 \\ -2\beta^2 + \gamma^2 & \alpha^2 - \beta^2 - \gamma^2 & \alpha^2 + \beta^2 + \gamma^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Equating the corresponding elements, we have

$$4\beta^2 + \gamma^2 = 1 \quad \dots(i)$$

$$2\beta^2 - \gamma^2 = 0 \quad \dots(ii)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 1 \quad \dots(iii)$$

$$\text{From (i) and (ii), } 6\beta^2 = 1 \quad \therefore \beta^2 = \frac{1}{6} \text{ and } \gamma^2 = \frac{1}{3}$$

$$\text{From (iii) } \alpha^2 = 1 - \beta^2 - \gamma^2 = 1 - \frac{1}{6} - \frac{1}{3} = \frac{1}{2}$$

$$\text{Hence, } \alpha = \pm \frac{1}{\sqrt{2}}, \beta = \pm \frac{1}{\sqrt{6}} \text{ and } \gamma = \pm \frac{1}{\sqrt{3}}$$

Ans.



(i) If $A = \begin{bmatrix} 4 & 2 & -5 \\ 1 & 0 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & -7 & 0 \\ -1 & 2 & 5 \\ 1 & 0 & 3 \end{bmatrix}$, then show that $(AB)^T = B^T \cdot A^T$.

(ii) If $A = \begin{bmatrix} 2 & 3 & -4 \\ -1 & 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ 3 & 4 \\ -5 & -6 \end{bmatrix}$, then find $A + B^T$.

(iii) If $A = \begin{bmatrix} 9 & -3 & 6 \\ 8 & \frac{1}{2} & 7 \\ -1 & 0 & 0 \end{bmatrix}$, then, show that $(A^T)^T = A$.

(iv) Show that the matrix $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$ is an orthogonal matrix.

Symmetric and Skew Symmetric Matrix

(a) Symmetric Matrix

A square matrix $A = [a_{ij}]$ is said to be, symmetric if, $a_{ij} = a_{ji} \forall i$ and j (conjugate elements are equal).

Hence for symmetric matrix $A = A^T$.

Note: Maximum number of distinct entries in any symmetric matrix of order n is $\frac{n(n+1)}{2}$.

(b) Skew Symmetric Matrix

Square matrix $A = [a_{ij}]$ is said to be skew symmetric if $a_{ij} = -a_{ji} \forall i$ and j (the pair of conjugate elements are additive inverse of each other). For a skew symmetric matrix $A = -A^T$.

- Note:** (i) If A is skew symmetric, then $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0 \forall i$. Thus, the diagonal elements of a skew square matrix are all zero, but not the converse.
- (ii) The determinant value of odd order skew symmetric matrix is zero.

(c) Properties of Symmetric and Skew Symmetric Matrix

- (i) A is symmetric if $A^T = A$ and A is skew symmetric if $A^T = -A$
- (ii) Let A be any square matrix then, $A + A^T$ is a symmetric matrix and $A - A^T$ is a skew symmetric matrix.
- (iii) The sum of two symmetric matrix is a symmetric matrix and the sum of two skew symmetric matrix is a skew symmetric matrix.
- (iv) If A and B are symmetric matrices then,
 (1) $AB + BA$ is a symmetric matrix
 (2) $AB - BA$ is a skew symmetric matrix.
- (v) Every square matrix can be uniquely expressed as a sum or difference of a symmetric and a skew symmetric matrix.

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{symmetric}} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{skew symmetric}} \text{ and } A = \frac{1}{2}(A^T + A) - \frac{1}{2}(A^T - A)$$

**Example 8:**

If A is symmetric as well as skew symmetric matrix, then A is-

- (A) diagonal matrix (B) null matrix
(C) triangular matrix (D) None of these

Solution:

Let $A = [a_{ij}]$ Since A is skew symmetric $a_{ij} = -a_{ji}$

for $i = j$, $a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$

for $i \neq j$, $a_{ij} = -a_{ji}$ [$\because$ A is skew symmetric] and $a_{ij} = a_{ji}$ [$\because$ A is symmetric]

$\therefore a_{ij} = 0$ for all $i \neq j$

so, $a_{ij} = 0$ for all 'i' and 'j' i.e. A is null matrix.

Ans. (B)**Concept Builders - 6**

(i) If $A = \begin{bmatrix} -2 & -1 & 1 \\ -1 & 7 & 4 \\ 1 & -x & -3 \end{bmatrix}$ be symmetric matrix then find the value of x.

(ii) Express matrix $A = \begin{bmatrix} 2 & 5 & 7 \\ 9 & -7 & 2 \\ 1 & -1 & 0 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

Adjoint of a Square Matrix

Let $A = [a_{ij}]$ be a square matrix of order n and let C_{ij} be cofactor of a_{ij} in A then the adjoint of A, denoted by **adj A**, is defined as the transpose of the cofactor matrix.

$$\text{Then, adj } A = [C_{ij}]^T \Rightarrow \text{adj } A = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{23} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}^T$$

Theorem

$$A (\text{adj. } A) = (\text{adj. } A) \cdot A = |A| I_n.$$

Proof:

$$A \cdot (\text{adj } A) = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \begin{pmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{pmatrix}$$

$$\begin{pmatrix} |A| & 0 & 0 \\ 0 & |A| & 0 \\ 0 & 0 & |A| \end{pmatrix} = |A| \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow A \cdot (\text{Adj. } A) = |A| I$$

(whatever may be the value only |A| will come out as a common element)

If $|A| \neq 0$, then $\frac{A \cdot (\text{adj. } A)}{|A|} = I = \text{unit matrix of the same order as that of } A$



Properties of Adjoint Matrix

If **A** be a square matrix of order **n**, then

- (i) $|\text{adj } A| = |A|^{n-1}$
- (ii) $\text{adj}(\text{adj } A) = |A|^{n-2} A$, where $|A| \neq 0$
- (iii) $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$, where $|A| \neq 0$
- (iv) $\text{adj}(AB) = (\text{adj } B)(\text{adj } A)$
- (v) $\text{adj}(KA) = K^{n-1}(\text{adj } A)$, K is a scalar
- (vi) $\text{adj } A^T = (\text{adj } A)^T$

Method to Find Adjoint of a 2×2 Square Matrix, Directly

Let **A** be a 2×2 square matrix. In order to find the adjoint simply interchange the diagonal elements and reverse the sign of off diagonal elements (rest of the elements).

e.g., If $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \Rightarrow \text{adj } A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$

Example 9:

If $A = \begin{bmatrix} 1 & 3 & 5 \\ 3 & 5 & 1 \\ 5 & 1 & 3 \end{bmatrix}$, then $\text{adj } A$ is equal to-

(A) $\begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$

(B) $\begin{bmatrix} -14 & 4 & 22 \\ 4 & 22 & -14 \\ 22 & -14 & 4 \end{bmatrix}$

(C) $\begin{bmatrix} 14 & 4 & -22 \\ 4 & -22 & -14 \\ -22 & -14 & -4 \end{bmatrix}$

(D) None of these

Solution:

$$\text{adj } A = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}^T = \begin{bmatrix} 14 & -4 & -22 \\ -4 & -22 & 14 \\ -22 & 14 & -4 \end{bmatrix}$$

Ans. (A)

Example 10:

If $A = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix}$, then $\text{adj}(\text{adj } A)$ is equal to-

(A) $8 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(B) $16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(C) $64 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

(D) None of these

Solution:

$$|A| = \begin{vmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{vmatrix} = 8$$

$$\text{Now } \text{adj}(\text{adj } A) = |A|^{3-2} A$$

$$= 8 \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 2 & 2 \end{bmatrix} = 16 \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Ans. (B)



- (i) For any 2×2 matrix, if $A(\text{Adj } A) = \begin{bmatrix} 15 & 0 \\ 0 & 15 \end{bmatrix}$, then $|A|$ is equal-
 (A) 20 (B) 625 (C) 15 (D) 0
- (ii) Which of the following is/are incorrect?
 (A) Adjoint of a symmetric matrix is skew symmetric matrix.
 (B) Adjoint of a diagonal matrix is a diagonal matrix.
 (C) $A(\text{Adj } A) = (\text{Adj } A) A = |A| I$
 (D) Adjoint of a unit matrix is a diagonal matrix
- (iii) If A be a square matrix of the order 5 and $B = \text{Adj}(A)$ then find $\text{Adj}(5A)$.
- (iv) If A be a square matrix of order 4 and $|A| = 3$ then find $\text{adj}(\text{adj } A)$.

Inverse of a Matrix (Reciprocal Matrix)

A square matrix A said to be invertible if and only if it is non-singular (i.e. $|A| \neq 0$) and there exists a matrix B such that, $AB = I = BA$.

B is called the **inverse** (reciprocal) of A and is denoted by A^{-1} . Thus

$$A^{-1} = B \Leftrightarrow AB = I = BA$$

$$\text{We have, } A(\text{adj } A) = |A| I_n$$

$$A^{-1} \cdot A(\text{adj } A) = A^{-1} I_n |A|$$

$$I_n(\text{adj } A) = A^{-1} |A| I_n$$

$$\therefore A^{-1} = \frac{(\text{adj } A)}{|A|}$$

Note: The necessary and sufficient condition for a square matrix A to be invertible is that $|A| \neq 0$

Properties of Inverse

- (i) If A and B are invertible matrices of the same order, then $(AB)^{-1} = B^{-1}A^{-1}$.

Note: If $A_1, A_2, \dots, A_n$ are all invertible square matrices of order n

$$\text{then } (A_1 A_2 \dots A_n)^{-1} = A_n^{-1} A_{n-1}^{-1} \dots A_2^{-1} A_1^{-1}$$

- (ii) If A be an invertible matrix, then A^T is also invertible and $(A^T)^{-1} = (A^{-1})^T$.

- (iii) If A is invertible,

$$(a) (A^{-1})^{-1} = A$$

$$(b) (A^k)^{-1} = (A^{-1})^k = A^{-k}; k \in \mathbb{N}$$

- (iv) If A is non-singular matrix, then $|A^{-1}| = |A|^{-1}$

- (v) If idempotent matrix is invertible then its inverse will be identity matrix.

- (vi) A nilpotent matrix will not be invertible because its determinant value is zero.

- (vii) Orthogonal matrix A is always invertible and $A^{-1} = A^T$.

- (viii) $A = A^{-1}$ for an involutory matrix.



Cancellation Law

Let A, B, C be square matrices of the same order 'n'.

If A is a non-singular matrix, then

- (a) $AB = AC \Rightarrow B = C$ (Left cancellation law)
 (b) $BA = CA \Rightarrow B = C$ (Right cancellation law)

Note that these cancellation laws hold only if the matrix 'A' is **non-singular** (i.e. $|A| \neq 0$).

Example 11:

Prove that if A is non-singular matrix such that A is symmetric then A^{-1} is also symmetric.

Solution:

$$\begin{aligned} A^T &= A && [\because A \text{ is a symmetric matrix}] \\ (A^T)^{-1} &= A^{-1} && [\text{Since } A \text{ is non-singular matrix}] \\ \Rightarrow (A^{-1})^T &= A^{-1} && \text{Thus } A^{-1} \text{ is a symmetric matrix.} \end{aligned}$$

Example 12:

$\begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan \theta / 2 \\ -\tan \theta / 2 & 1 \end{bmatrix}^{-1}$ is equal to-

- (A) $\begin{bmatrix} \sin \theta & -\cos \theta \\ \cos \theta & \sin \theta \end{bmatrix}$ (B) $\begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ (C) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ (D) None of these

Solution:

$$\begin{aligned} \begin{bmatrix} 1 & \tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix}^{-1} &= \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \\ \therefore \text{Product} &= \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\tan \theta / 2 \\ \tan \theta / 2 & 1 \end{bmatrix} \\ &= \frac{1}{\sec^2 \theta / 2} \begin{bmatrix} 1 - \tan^2 \theta / 2 & -2 \tan \theta / 2 \\ 2 \tan \theta / 2 & 1 - \tan^2 \theta / 2 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta / 2 - \sin^2 \theta / 2 & -2 \sin \theta / 2 \cos \theta / 2 \\ 2 \sin \theta / 2 \cos \theta / 2 & \cos^2 \theta / 2 - \sin^2 \theta / 2 \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \end{aligned}$$

Ans. (C)

Example 13:

If $A = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$ and $M = AB$, then M^{-1} is equal to-

- (A) $\begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1/3 & 1/3 \\ -1/3 & 1/6 \end{bmatrix}$ (C) $\begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$ (D) $\begin{bmatrix} 1/3 & -1/3 \\ -1/3 & 1/6 \end{bmatrix}$

Solution:

$$M = \begin{bmatrix} 0 & -1 & 2 \\ 2 & -2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ -2 & 2 \end{bmatrix}$$

$$|M| = 6, \text{adj } M = \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix}$$

$$\therefore M^{-1} = \frac{1}{6} \begin{bmatrix} 2 & -2 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1/3 & -1/3 \\ 1/3 & 1/6 \end{bmatrix}$$

Ans. (C)



- (i) If 'A' is a square matrix such that $A^2 = I$ then A^{-1} is equal to-
 (A) $A + I$ (B) A (C) 0 (D) $2A$
- (ii) If 'A' is an orthogonal matrix, then A^{-1} equal-
 (A) A (B) A^T (C) A^2 (D) None of these
- (iii) If $A = \begin{bmatrix} 3 & 2 \\ 0 & 1 \end{bmatrix}$, then $(A^{-1})^3$ is equal to-
 (A) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & 27 \end{bmatrix}$ (B) $\frac{1}{27} \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}$ (C) $\frac{1}{27} \begin{bmatrix} 1 & -26 \\ 0 & -27 \end{bmatrix}$ (D) $\frac{1}{27} \begin{bmatrix} -1 & -26 \\ 0 & -27 \end{bmatrix}$

Matrix Polynomial

If $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_nx^0$, then we define a matrix polynomial

$$f(A) = a_0A^n + a_1A^{n-1} + a_2A^{n-2} + \dots + a_nI^n.$$

Where A is the given square matrix. If $f(A)$ is the null matrix, then A is called the zero or root of the polynomial $f(x)$.

Characteristic Equation

Let A be a square matrix. Then the polynomial $|A - xI|$ is called as characteristic polynomial of A and the equation $|A - xI| = 0$ is called as characteristic equation of A. After solving the characteristic polynomial the values of 'x' are said to be characteristic roots of the polynomial.

- Note:** (i) Sum of the roots of the characteristic equation is equal to trace of the matrix.
 (ii) Product of the roots of the characteristic equation is equal to the determinant value.
 (iii) The degree of characteristic equation is same as the order of the matrix.

Example 14:

If $f(x) = x^2 - 3x + 3$ and $A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix}$ be a square matrix then prove that $f(A) = \mathbf{O}$.

Hence find A^4 .

Solution:

$$A^2 = A.A = \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix}$$

$$\text{Hence } A^2 - 3A + 3I = \begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - 3 \begin{bmatrix} 2 & 1 \\ -1 & 1 \end{bmatrix} + 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbf{O}$$



Alter

$$\because |A - XI| = 0 \Rightarrow \begin{vmatrix} 2-x & 1 \\ -1 & 1-x \end{vmatrix} = 0$$

$$\Rightarrow (2-x)(1-x) + 1 = 0 \Rightarrow x^2 - 3x + 3 = 0 \text{ (characteristic polynomial)}$$

By Cayley-Hamilton Theorem $A^2 - 3A + 3I = \mathbf{O}$. Hence proved.

$$\text{Now } A^2 = 3A - 3I$$

Squaring on both the sides

$$A^4 = 9(A^2 - 2A + I)$$

$$= 9 \left(\begin{bmatrix} 3 & 3 \\ -3 & 0 \end{bmatrix} - 2 \begin{bmatrix} 4 & 2 \\ -2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) = 9 \begin{bmatrix} 3-4+1 & 3-2 \\ -3+2 & -2+1 \end{bmatrix}$$

$$= 9 \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 9 \\ -9 & -9 \end{bmatrix}$$

Cayley-Hamilton Theorem

Every square matrix A satisfy its characteristic equation

i.e., $a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = 0$ is the characteristic equation of A, then

$$a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

Note: This theorem is helpful to find the inverse of any non-singular square matrix.

$$\text{i.e. } a_0A^n + a_1A^{n-1} + \dots + a_{n-1}A + a_nI = \mathbf{O}$$

On multiplying by A^{-1} on both the sides of above equation, we get

$$A^{-1} = -\frac{1}{a_n}(a_0A^{n-1} + a_1A^{n-2} + \dots + a_{n-1}I)$$

Example 15:

$$\text{If } A = \begin{bmatrix} 1 & 2 & 0 \\ 2 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \text{ show that } 5A^{-1} = A^2 + A - 5I$$

Solution:

We have the characteristic equation of A.

$$|A - xI| = 0$$

$$\text{i.e. } \begin{vmatrix} 1-x & 2 & 0 \\ 2 & -1-x & 0 \\ 0 & 0 & -1-x \end{vmatrix} = 0$$

$$\text{i.e. } x^3 + x^2 - 5x - 5 = 0$$

Using Cayley - Hamilton theorem

$$A^3 + A^2 - 5A - 5I = \mathbf{O}$$

$$\Rightarrow 5I = A^3 + A^2 - 5A$$

$$\text{Multiplying by } A^{-1}, \text{ we get } 5A^{-1} = A^2 + A - 5I$$



- (i) Determine the characteristic roots of the matrix A. Hence find the trace the determinant value of A.

Where $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ and also prove that $A^3 - 18A^2 + 45A = \mathbf{O}$.

System of Equation and Criteria for Consistency

Gauss-Jordan Method

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

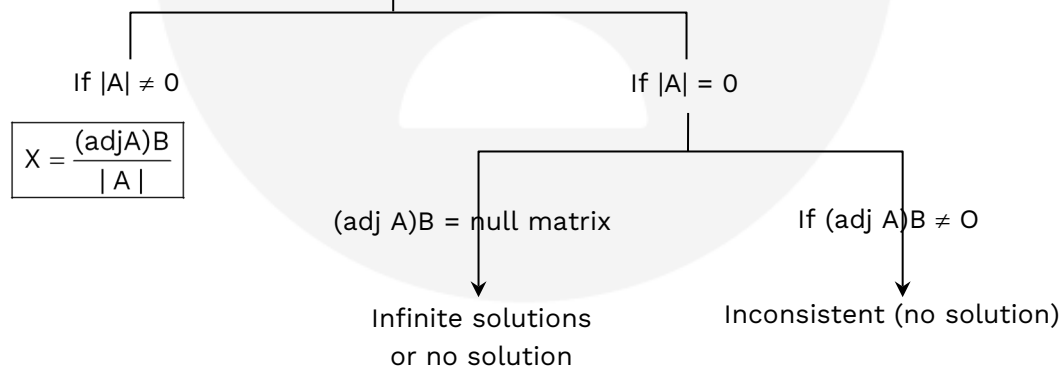
$$\Rightarrow \begin{bmatrix} a_1x + b_1y + c_1z \\ a_2x + b_2y + c_2z \\ a_3x + b_3y + c_3z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix} \Rightarrow \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$$

$$\Rightarrow AX = B \quad \dots(i)$$

Multiplying **adj A** on both the sides of (i)

$$\Rightarrow (\text{adj}A) AX = (\text{adj}A)B \Rightarrow |A|X = (\text{adj}A)B$$

$$|A|X = (\text{adj}A)B$$



Example 16:

Solve the system of equations $x + y + z = 6$
 $x - y + z = 2$ using matrix method.
 $2x + y - z = 1$

Solution:

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ & $B = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$

Then the system is $AX = B$.



$|A| = 6$, hence A is nonsingular,

$$\text{Cofactor } A = \begin{bmatrix} 0 & 3 & 3 \\ 2 & -3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$$

$$\text{adj } A = \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{|A|} \text{adj} A = \frac{1}{6} \begin{bmatrix} 0 & 2 & 2 \\ 3 & -3 & 0 \\ 3 & 1 & -2 \end{bmatrix} = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix}$$

$$X = A^{-1}B = \begin{bmatrix} 0 & 1/3 & 1/3 \\ 1/2 & -1/2 & 0 \\ 1/2 & 1/6 & -1/3 \end{bmatrix} \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix} \quad \text{i.e.} \quad \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$\Rightarrow x = 1, y = 2, z = 3$$

Ans.

Concept Builders - 10



- (i) The system of equations $x + 2y - 3z = 1$, $x - y + 4z = 0$, $2x + y + z = 1$ has-
- (A) only two solutions (B) only one solution
- (C) no solution (D) infinitely many solutions
- (ii) The system of equations $x + y + z = 8$, $x - y + 2z = 6$, $3x + 5y - 7z = 14$ has-
- (A) Unique solution (B) infinite number of solutions
- (C) no solution (D) none of these



ANSWER KEY FOR CONCEPT BUILDER

1. (i) $\begin{bmatrix} 3 & 5 & 7 \\ 4 & 6 & 8 \end{bmatrix}$ (ii) 6
(iii) 30 (iv) $x = 2, y = -1, a = 1, b = -3$
2. (i) $\begin{bmatrix} -7 & -10 & -7 \\ -2 & 6 & 3 \end{bmatrix}$ and 2×3 (ii) $\begin{bmatrix} -8 & -9 \\ -7/2 & -8 \\ -1 & 1 \end{bmatrix}$
3. (iii) $x = -2$
5. (ii) $\begin{bmatrix} 1 & 6 & -9 \\ 1 & 6 & -3 \end{bmatrix}$
6. (i) -4 (ii) $\begin{bmatrix} 2 & 7 & 4 \\ 7 & -7 & \frac{1}{2} \\ 4 & \frac{1}{2} & 0 \end{bmatrix} + \begin{bmatrix} 0 & -2 & 3 \\ 2 & 0 & \frac{3}{2} \\ -3 & -\frac{3}{2} & 0 \end{bmatrix}$
7. (i) C (ii) A (iii) 625 B (iv) 9A
8. (i) B (ii) B (iii) A
9. (i) $\lambda = 0, 3$ and 15 $\text{tr}(A) = 18, |A| = 0$
10. (i) D (ii) A

Objective Exercise - I

1. If $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, then
 (A) $x = 3, y = 7, z = 1, w = 14$ (B) $x = 3, y = -5, z = -1, w = -4$
 (C) $x = 3, y = 6, z = 2, w = 7$ (D) None of these
2. Identify the incorrect statement in respect of two square matrices A and B conformable for sum and product-
 (A) $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$ (B) $\text{tr}(\alpha A) = \alpha \text{tr}(A), \alpha \in \mathbb{R}$
 (C) $\text{tr}(A^T) = \text{tr}(A)$ (D) $\text{tr}(AB) \neq \text{tr}(BA)$
3. A and B are two given matrices such that the order of A is 3×4 , if $A'B$ and BA' are both defined then
 (A) order of B' is 3×4 (B) order of $B'A$ is 4×4
 (C) order of $B'A$ is 3×3 (D) $B'A$ is undefined
4. Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$, then $\text{Tr}(A) - \text{Tr}(B)$ has the value equal to
 (A) 0 (B) 1 (C) 2 (D) none
5. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $J = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then $B =$
 (A) $I\cos\theta + J\sin\theta$ (B) $I\cos\theta - J\sin\theta$ (C) $I\sin\theta + J\cos\theta$ (D) $-I\cos\theta + J\sin\theta$
6. The matrix $A^2 + 4A - 5I$, where I is identity matrix and $A = \begin{bmatrix} 1 & 2 \\ 4 & -3 \end{bmatrix}$ equals:
 (A) $32 \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ (B) $4 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$ (C) $4 \begin{bmatrix} 0 & -1 \\ 2 & 2 \end{bmatrix}$ (D) $32 \begin{bmatrix} 2 & 1 \\ 2 & 0 \end{bmatrix}$
7. If $A = \text{diag}(2, -1, 3)$, $B = \text{diag}(-1, 3, 2)$, then $A^2B =$
 (A) $\text{diag}(5, 4, 11)$ (B) $\text{diag}(-4, 3, 18)$ (C) $\text{diag}(3, 1, 8)$ (D) B
8. $A = \begin{bmatrix} 0 & 1 \\ 2 & 0 \end{bmatrix}$ and $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 31 \\ 62 \end{bmatrix}$.
 (Where I is the (2×2) identity matrix), then the product of all elements of matrix V is
 (A) 2 (B) 1 (C) 3 (D) -2
9. If $M = \begin{bmatrix} 0 & 2 \\ 5 & 0 \end{bmatrix}$ and $N = \begin{bmatrix} 0 & 5 \\ 2 & 0 \end{bmatrix}$, then M^{2011} is -
 (A) $10^{1005}M$ (B) $10^{1005}N$ (C) $10^{2010}M$ (D) $10^{2011}M$
10. If $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $A^2 - kA - I_2 = 0$, then value of k is-
 (A) 4 (B) 2 (C) 1 (D) -4



11. If the product of n matrices $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \dots \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$ is equal to the matrix $\begin{bmatrix} 1 & 378 \\ 0 & 1 \end{bmatrix}$ then the value of n is equal to -
(A) 26 (B) 27 (C) 377 (D) 378
12. If p, q, r are 3 real number satisfying the matrix equation, $[p \ q \ r] \begin{bmatrix} 3 & 4 & 1 \\ 3 & 2 & 3 \\ 2 & 0 & 2 \end{bmatrix} = [3 \ 0 \ 1]$, then $2p + q - r$ equals: -
(A) -1 (B) 4 (C) -3 (D) 2
13. Which of the following is an orthogonal matrix -
(A) $\begin{bmatrix} 6/7 & 2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ 3/7 & -6/7 & 2/7 \end{bmatrix}$ (B) $\begin{bmatrix} 6/7 & 2/7 & 3/7 \\ 2/7 & -3/7 & 6/7 \\ 3/7 & 6/7 & -2/7 \end{bmatrix}$
(C) $\begin{bmatrix} -6/7 & -2/7 & -3/7 \\ 2/7 & 3/7 & 6/7 \\ -3/7 & 6/7 & 2/7 \end{bmatrix}$ (D) $\begin{bmatrix} 6/7 & -2/7 & 3/7 \\ 2/7 & 2/7 & -3/7 \\ -6/7 & 2/7 & 3/7 \end{bmatrix}$
14. Let three matrices are $A = \begin{bmatrix} 2 & 1 \\ 4 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 3 & 4 \\ 2 & 3 \end{bmatrix}$ and $C = \begin{bmatrix} 3 & -4 \\ -2 & 3 \end{bmatrix}$, then $t_r(A) + t_r\left(\frac{ABC}{2}\right) + t_r\left(\frac{A(BC)^2}{4}\right) + t_r\left(\frac{A(BC)^3}{8}\right) + \dots + \infty$ is equal to -
(A) 6 (B) 9 (C) 12 (D) none
15. For a matrix $A = \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$, the value of $\prod_{r=1}^{50} \begin{bmatrix} 1 & 2r-1 \\ 0 & 1 \end{bmatrix}$ is equal to -
(A) $\begin{bmatrix} 1 & 100 \\ 0 & 1 \end{bmatrix}$ (B) $\begin{bmatrix} 1 & 4950 \\ 0 & 1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 5050 \\ 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 2500 \\ 0 & 1 \end{bmatrix}$
16. Matrix $A = \begin{bmatrix} x & 3 & 2 \\ 1 & y & 4 \\ 2 & 2 & z \end{bmatrix}$, if $xyz = 60$ and $8x + 4y + 3z = 20$, then $A(\text{adj } A)$ is equal to:-
(A) $\begin{bmatrix} 64 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 64 \end{bmatrix}$ (B) $\begin{bmatrix} 88 & 0 & 0 \\ 0 & 88 & 0 \\ 0 & 0 & 88 \end{bmatrix}$
(C) $\begin{bmatrix} 68 & 0 & 0 \\ 0 & 68 & 0 \\ 0 & 0 & 68 \end{bmatrix}$ (D) $\begin{bmatrix} 34 & 0 & 0 \\ 0 & 34 & 0 \\ 0 & 0 & 34 \end{bmatrix}$
17. If A and B are symmetric matrices, then ABA is -
(A) symmetric matrix (B) skew symmetric matrix
(C) diagonal matrix (D) scalar matrix



18. Let $A = \begin{pmatrix} 0 & \sin \alpha & \sin \alpha \sin \beta \\ -\sin \alpha & 0 & \cos \alpha \cos \beta \\ -\sin \alpha \sin \beta & -\cos \alpha \cos \beta & 0 \end{pmatrix}$, then-
- (A) $|A|$ is independent of α and β (B) A^{-1} depends only on α
(C) A^{-1} depends only on β (D) none of these
19. Consider a matrix $A(\theta) = \begin{bmatrix} \sin \theta & \cos \theta \\ -\cos \theta & \sin \theta \end{bmatrix}$ then,
- (A) $A(\theta)$ is symmetric (B) $A(\theta)$ is skew symmetric
(C) $A^{-1}(\theta) = A(\pi - \theta)$ (D) $A^2(\theta) = A\left(\frac{\pi}{2} - 2\theta\right)$
20. Number of real values of λ for which the matrix $A = \begin{bmatrix} \lambda - 1 & \lambda & \lambda + 1 \\ 2 & -1 & 3 \\ \lambda + 3 & \lambda - 2 & \lambda + 7 \end{bmatrix}$ has no inverse
- (A) 0 (B) 1 (C) 2 (D) infinite
21. If $A = [a_{ij}]_{2 \times 2}$ where $a_{ij} = \begin{cases} i+j & i \neq j \\ i^2 - 2j & i = j \end{cases}$, then A^{-1} is equal to -
- (A) $\frac{1}{9} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$ (B) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ 3 & -1 \end{bmatrix}$ (C) $\frac{1}{9} \begin{bmatrix} 0 & -3 \\ -3 & -1 \end{bmatrix}$ (D) $\frac{1}{3} \begin{bmatrix} 0 & 3 \\ 3 & 1 \end{bmatrix}$
22. If $A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & a & 1 \end{bmatrix}$, $A^{-1} = \begin{bmatrix} 1/2 & -1/2 & 1/2 \\ -4 & 3 & c \\ 5/2 & -3/2 & 1/2 \end{bmatrix}$, then -
- (A) $a = 1, c = -1$ (B) $a = 2, c = -\frac{1}{2}$ (C) $a = -1, c = 1$ (D) $a = \frac{1}{2}, c = \frac{1}{2}$
23. Let the matrix A and B be define as $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 1 \\ 7 & 3 \end{bmatrix}$ then the value of $\text{Det. } (2A^9B^{-1})$, is-
- (A) 2 (B) 1 (C) -1 (D) -2

ANSWER KEY

- | | | | | | | | | | | | | | |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 1. | (A) | 2. | (D) | 3. | (B) | 4. | (C) | 5. | (A) | 6. | (B) | 7. | (B) |
| 8. | (A) | 9. | (A) | 10. | (A) | 11. | (B) | 12. | (C) | 13. | (A) | 14. | (A) |
| 15. | (D) | 16. | (C) | 17. | (A) | 18. | (A) | 19. | (C) | 20. | (D) | 21. | (A) |
| 22. | (A) | 23. | (D) | | | | | | | | | | |

Objective Exercise - II

1. Let P and Q are orthogonal matrices of order 3×3 such that $A = PQ^T$ and $B = QP^T$, then the value of $\text{tr}(AB)$ is equal to-
 (A) 1 (B) 3 (C) 5 (D) 7
2. Let $S = \left\{ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} : a_{ij} \in \{0, 1, 2\}, a_{11} = a_{22} \right\}$. Then the number of non-singular matrices in the set S is-
 (A) 24 (B) 10 (C) 20 (D) 27
3. If $A = \begin{bmatrix} \cos^2 \alpha & \sin \alpha \cos \alpha \\ \sin \alpha \cos \alpha & \sin^2 \alpha \end{bmatrix}$; $B = \begin{bmatrix} \cos^2 \beta & \sin \beta \cos \beta \\ \sin \beta \cos \beta & \sin^2 \beta \end{bmatrix}$ are such that, AB is a null matrix, then which of the following should necessarily be an odd integral multiple of $\frac{\pi}{2}$ -
 (A) α (B) β (C) $\alpha - \beta$ (D) $\alpha + \beta$
4. A is an involutory matrix given by $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ then the inverse of $\frac{A}{2}$ will be-
 (A) 2A (B) $\frac{A^{-1}}{2}$ (C) $\frac{A}{2}$ (D) A^2
5. If A and B are invertible matrices, which one of the following statements is/are incorrect-
 (A) $\text{Adj}(A) = |A|A^{-1}$ (B) $\det(A^{-1}) = |\det(A)|^{-1}$
 (C) $(A + B)^{-1} = B^{-1} + A^{-1}$ (D) $(AB)^{-1} = B^{-1}A^{-1}$
6. Let $A = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 5 & 2 \\ 1 & 2 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -x & -y & z \\ 0 & y & 2z \\ x & -y & z \end{bmatrix}$ where $x, y, z \in \mathbb{R}$. If $B^T A B = \begin{bmatrix} 8 & 0 & 0 \\ 0 & 27 & 0 \\ 0 & 0 & 42 \end{bmatrix}$ then the number of ordered triplet (x, y, z) is-
 (A) 2 (B) 6 (C) 8 (D) 9
7. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{bmatrix}$ and $10B = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & \alpha \\ 1 & -2 & 3 \end{bmatrix}$. If B is the inverse of matrix A, then α is-
 (A) -2 (B) -1 (C) 2 (D) 5
8. $A = \begin{bmatrix} 1 & \tan x \\ -\tan x & 1 \end{bmatrix}$ then let us define a function $f(x) = \det. (A^T A^{-1})$ then which of the following can not be the value of $\underbrace{f(f(f(\dots f(x))))}_{n \text{ times}}$ is ($n \geq 2$)
 (A) $f^n(x)$ (B) 1 (C) $f^{n-1}(x)$ (D) $nf(x)$



One or More Than One Correct Type Questions

9. Let $\theta = \frac{\pi}{5}$, $X = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, O is null matrix and I is an identity matrix of order 2×2 , and if $I + X + X^2 + \dots + X^n = O$, then n can be
(A) 9 (B) 19 (C) 4 (D) 29
10. Let A, B, C, D be real matrices such that $A^T = BCD$; $B^T = CDA$; $C^T = DAB$ and $D^T = ABC$ for the matrix $M = ABCD$, then find M^{2016} ?
(A) M (B) M^2 (C) M^3 (D) M^4
11. Let A and B be two 2×2 matrix with real entries. If $AB = O$ and $\text{tr}(A) = \text{tr}(B) = 0$ then
(A) A and B are commutative w.r.t. operation of multiplication.
(B) A and B are not commutative w.r.t. operation of multiplication.
(C) A and B are both null matrices.
(D) $BA = O$
12. If A and B are square matrices of order 3, then the true statement is/are (where I is unit matrix).
(A) $\det(-A) = -\det A$
(B) If AB is singular then atleast one of A or B is singular
(C) $\det(A + I) = 1 + \det A$
(D) $\det(2A) = 2^3 \det A$
13. Let $\det(\text{adj}(\text{adj}A)) = 14^4$ where $A = \begin{bmatrix} x & 2 & -1 \\ -1 & 1 & 2 \\ 2 & -1 & 1 \end{bmatrix}$, $x \neq -\frac{11}{3}$, then
(A) $x = 1$ (B) $\det(2A) = 112$ (C) $x = 2$ (D) $\det(2A) = 256$
14. Let $A^{-1} = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 2 & 1 \\ 2 & 1 & -1 \end{bmatrix}$, then-
(A) $7 | A | = \frac{1}{2}$ (B) $|\text{adj } A| = \frac{1}{196}$
(C) $\text{trace}(\text{adj } A) = -\frac{1}{7}$ (D) Matrix A is symmetric matrix
15. If $A = \begin{bmatrix} 3 & 0 & 0 \\ 2 & 2 & 0 \\ 4 & 5 & 3 \end{bmatrix}$, then which of the following is(are) true?
(trace of A denotes sum of principal diagonal elements of A)
(A) A is invertible (B) $\text{trace}(\text{adj}(\text{adj}(A))) = 144$
(C) $\text{trace}(\text{adj}(\text{adj}(A))) = 8$ (D) $|\text{adj } A|$ is less than 400



16. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & -1 \\ 3 & 0 & k \end{bmatrix}$ and $f(x) = x^3 - 2x^2 - \alpha x + \beta = 0$. If A satisfies $f(x) = 0$, then-
- (A) $k = 1, \alpha = 14$ (B) $\alpha = 14, \beta = 22$ (C) $k = -1, \beta = 22$ (D) $\alpha = -14, \beta = -22$
17. If A and B are 3×3 matrices and $|A| \neq 0$, then which of the following are true?
- (A) $|AB| = 0 \Rightarrow |B| = 0$ (B) $|AB| = 0 \Rightarrow B = 0$
(C) $|A^{-1}| = |A|^{-1}$ (D) $|A + A| = 2|A|$
18. If D_1 and D_2 are two 3×3 diagonal matrices where none of the diagonal element is zero, then -
- (A) $D_1 D_2$ is a diagonal matrix (B) $D_1 D_2 = D_2 D_1$
(C) $D_1^2 + D_2^2$ is a diagonal matrix (D) none of these
19. If A and B are two 3×3 matrices such that their product AB is a null matrix then
- (A) $\det. A \neq 0 \Rightarrow B$ must be a null matrix.
(B) $\det. B \neq 0 \Rightarrow A$ must be a null matrix
(C) If none of A and B are null matrices then atleast one of the two matrices must be singular.
(D) If neither $\det. A$ nor $\det. B$ is zero then given statement is not possible.
20. Let $A = a_{ij}$ be a matrix of order 3 where $a_{ij} = \begin{cases} x & \text{if } i = j, \\ 1 & \text{if } |i - j| = 1 \\ 0 & \text{otherwise} \end{cases} \quad x \in \mathbb{R}$, then which of the following hold(s) good?
- (A) for $x = 2$, A is a diagonal matrix
(B) A is a symmetric matrix
(C) for $x = 2$, $\det A$ has the value equal to 6
(D) Let $f(x) = \det A$, then the function $f(x)$ has both the maxima and minima.

Paragraph Type Questions

Paragraph for question nos. 21 to 22

Consider the system $AX = B$, where $A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ 3 & 2 & 2 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $B = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}$.

21. Sum of elements of $(\text{adj}A) B$ is -
- (A) -1 (B) 2 (C) -2 (D) -4
22. Value of $\text{tr}(XB^T)$ is (where $\text{tr}(A)$ denotes trace of matrix A)-
- (A) 0 (B) 1 (C) 2 (D) 3



Paragraph for question nos. 23 to 25

If A is a symmetric and B skew symmetric matrix and A + B is nonsingular and $C = (A + B)^{-1}(A - B)$ then

23. $C^T(A + B)C =$

(A) A + B

(B) A - B

(C) A

(D) B

24. $C^T(A - B)C =$

(A) A + B

(B) A - B

(C) A

(D) B

25. C^TAC

(A) A + B

(B) A - B

(C) A

(D) B

ANSWERS KEY

- | | | | | | | | | | | | | | |
|------------|-------|------------|-------|------------|------|------------|-------|------------|--------|------------|------|------------|-------|
| 1. | (B) | 2. | (C) | 3. | (C) | 4. | (A) | 5. | (C) | 6. | (C) | 7. | (D) |
| 8. | (D) | 9. | (ABD) | 10. | (BD) | 11. | (AD) | 12. | (ABD) | 13. | (AB) | 14. | (BCD) |
| 15. | (ABD) | 16. | (BC) | 17. | (AC) | 18. | (ABC) | 19. | (ABCD) | 20. | (D) | 21. | (C) |
| 22. | (A) | 23. | (A) | 24. | (B) | 25. | (C) | | | | | | |

Subjective Exercise - I

1. Let $M = \begin{bmatrix} a & -360 \\ b & c \end{bmatrix}$, where a, b and c are integers. Find the smallest positive value of b such that $M^2 = \mathbf{O}$, where $\mathbf{O}$ denotes 2×2 null matrix.
2. Find the number of 2×2 matrix satisfying following conditions:
 (i) a_{ij} is 1 or -1 ; (ii) $a_{11}a_{21} + a_{12}a_{22} = 0$
3. Find the value of x and y that satisfy the equations $\begin{bmatrix} 3 & -2 \\ 3 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} y & y \\ x & x \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3y & 3y \\ 10 & 10 \end{bmatrix}$
4. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p \\ q \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$. Such that $AB = B$ and $a + d = 5050$. Find the value of $(ad - bc)$.
5. If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then show that $F(x) \cdot F(y) = F(x + y)$. Hence prove that $[F(x)]^{-1} = F(-x)$.
6. If $C = \begin{bmatrix} 1 & 4 & 6 \\ 7 & 2 & 5 \\ 9 & 8 & 3 \end{bmatrix} \begin{bmatrix} 0 & 2 & 3 \\ -2 & 0 & 4 \\ -3 & -4 & 0 \end{bmatrix} \begin{bmatrix} 1 & 7 & 9 \\ 4 & 2 & 8 \\ 6 & 5 & 3 \end{bmatrix}$, then trace of $C + C^3 + C^5 + \dots + C^{99}$ is
7. Define $A = \begin{bmatrix} 0 & 1 \\ 3 & 0 \end{bmatrix}$. Find a vertical vector V such that $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$ (where I is the 2×2 identity matrix).
8. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 - 6x^2 + 7x + 2$.
9. If the matrices $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ (a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the matrix which commutes with A is of the form $\begin{bmatrix} \alpha - \beta & 2\beta/3 \\ \beta & \alpha \end{bmatrix}$.
10. (i) Prove that $(\text{adj adj } A) = |A|^{n-2} A$
 (ii) Find the value of $|\text{adj adj adj } A|$ in terms of $|A|$



11. $A = \begin{pmatrix} 3 & a & -1 \\ 2 & 5 & c \\ b & 8 & 2 \end{pmatrix}$ is Symmetric and $B = \begin{pmatrix} d & 3 & a \\ b-a & e & -2b-c \\ -2 & 6 & -f \end{pmatrix}$ is skew symmetric, then find AB.

Is AB a symmetric, skew symmetric or neither of them. Justify your answer.

12. Express the matrix $\begin{bmatrix} 1 & 2 & 5 \\ 2 & 3 & -6 \\ -1 & 0 & 4 \end{bmatrix}$ as a sum of a lower triangular matrix and an upper triangular matrix with zero in its leading diagonal. Also express the matrix as a sum of a symmetric and a skew symmetric matrix.

13. Let $A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ are two matrices such that $AB = (AB)^{-1}$ and $AB \neq I$ (where I is an identity matrix of order 3×3). Find the value of $\text{Tr. } (AB + (AB)^2 + (AB)^3 + \dots + (AB)^{100})$, where $\text{Tr. } (A)$ denotes the trace of matrix A .

14. Let $A = \begin{bmatrix} 3x^2 \\ 1 \\ 6x \end{bmatrix}$, $B = [a \ b \ c]$ and $C = \begin{bmatrix} (x+2)^2 & 5x^2 & 2x \\ 5x^2 & 2x & (x+2)^2 \\ 2x & (x+2)^2 & 5x^2 \end{bmatrix}$ be three given matrices, where a, b, c and $x \in \mathbb{R}$. Given that $\text{tr}(AB) = \text{tr}(C) \ \forall x \in \mathbb{R}$, where $\text{tr}(A)$ denotes trace of A . Find the value of $(a + b + c)$

15. Determine the values of a and b for which the system $\begin{bmatrix} 3 & -2 & 1 \\ 5 & -8 & 9 \\ 2 & 1 & a \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} b \\ 3 \\ -1 \end{bmatrix}$

- (a) has a unique solution
- (b) has no solution and
- (c) has infinitely many solutions

16. If A and B are square matrices of order 3, where $|A| = -2$ and $|B| = 1$, then find $\left| (A^{-1}) \text{adj}(B^{-1}) \text{adj}(2A^{-1}) \right|$.

17. If $\left\{ \frac{1}{2}(A - A' + I) \right\}^{-1} = \frac{2}{\lambda} \begin{bmatrix} \lambda - 13 & -\frac{\lambda}{3} & \frac{\lambda}{3} \\ -17 & 10 & -1 \\ 7 & -11 & 5 \end{bmatrix}$ for $A = \begin{bmatrix} -2 & 3 & 4 \\ 5 & -4 & -3 \\ 7 & 2 & 9 \end{bmatrix}$, then λ is :

ANSWER KEY

1. 10

2. 8

3. $x = \frac{3}{2}, y = 2$

4. 5049

6. Zero

7. $V = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{11} \end{bmatrix}$

9. 1

10. (ii) $|A|^{(n-1)^3}$

11. AB is neither symmetric nor skew symmetric

12. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -1 & 0 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 2 & 5 \\ 0 & 0 & -6 \\ 0 & 0 & 0 \end{bmatrix}; \begin{bmatrix} 1 & 2 & 2 \\ 2 & 3 & -3 \\ 2 & -3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 3 \\ 0 & 0 & -3 \\ -3 & 3 & 0 \end{bmatrix}$

13. 100

14. 7

15. (i) $a \neq -3, b \in \mathbb{R}$ (ii) $a = -3$ and $b \neq 1/3$ (iii) $a = -3, b = 1/3$

16. -8

17. 39

Subjective Exercise - II

1. (a) A is a square matrix of order n.
 ℓ = maximum number of distinct entries if A is a triangular matrix
 m = maximum number of distinct entries if A is a diagonal matrix
 p = minimum number of zeroes if A is a triangular matrix
 If $\ell + 5 = p + 2m$, find the order of the matrix.
 (b) Let A be the set of all 3×3 skew symmetric matrices whose entries are either $-1, 0$ or
 If there are exactly three 0's, three 1's and three (-1) 's, then find the number of such matrices.

2. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then prove that value of f and g satisfying the matrix equation $A^2 + fA + gI = O$ are equal to $(- \text{tr}(A))$ and determinant of A respectively. Given a, b, c, d are non-zero real numbers and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}; O = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

3. Let A be the 2×2 matrices given by $A = [a_{ij}]$, where $a_{ij} \in \{0, 1, 2, 3, 4\}$
 such that $a_{11} + a_{12} + a_{21} + a_{22} = 4$
 (i) Find the number of matrices A such that the trace of A is equal to 4.
 (ii) Find the number of matrices A such that A is invertible.
 (iii) Find the absolute value of the difference between maximum value and minimum value of $\det(A)$.
 (iv) Find the number of matrices A such that A is either symmetric or skew-symmetric or both and $\det(A)$ is divisible by 2

4. Let X be the solution set of the equation
 $A^X = I$, where $A = \begin{bmatrix} 0 & 1 & -1 \\ 4 & -3 & 4 \\ 3 & -3 & 4 \end{bmatrix}$ and I is the unit matrix and $X \subset \mathbb{N}$ then the minimum value of
 $\sum_x (\cos^x \theta + \sin^x \theta)$, $\theta \in \mathbb{R}$ is:

5. If A is a diagonal matrix of order 3×3 is commutative with every square matrix of order 3×3 under multiplication and $\text{tr}(A) = 12$, then the value of $|A|$ is :

6. A, is a (3×3) diagonal matrix having integral entries such that $\det(A) = 120$, number of such matrices is $10n$. Then n is :

7. If A is a square matrix of order 3 and A' denotes transpose of matrix A, $A' A = I$ and $\det A = 1$, then $\det(A - I)$ must be equal to:



8. For the matrix $A = \begin{bmatrix} 4 & -4 & 5 \\ -2 & 3 & -3 \\ 3 & -3 & 4 \end{bmatrix}$ find A^{-2} .
9. (a) Given $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 1 \\ 2 & 3 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 3 \\ 3 & 4 \end{bmatrix}$. Find P such that $BPA = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$
- (b) Find the matrix A satisfying the matrix equation, $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} A \begin{bmatrix} 3 & 2 \\ 5 & -3 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 3 & -1 \end{bmatrix}$
10. Let A be a 3×3 matrix such that $a_{11} = a_{33} = 2$ and all the other $a_{ij} = 1$. Let $A^{-1} = xA^2 + yA + zI$, then find the value of $(x + y + z)$ where I is a unit matrix of order 3.
11. Given that $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 2 & 3 \\ 1 & -1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 2 & 1 & 1 \\ 2 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, $D = \begin{bmatrix} 10 \\ 13 \\ 9 \end{bmatrix}$ and that $CB = D$. Solve the matrix equation $Ax = B$.
12. Let $A = \begin{bmatrix} 1 & \frac{3}{2} \\ 1 & \frac{2}{2} \end{bmatrix}$, $B = \begin{bmatrix} 4 & -3 \\ -2 & 2 \end{bmatrix}$ and $C_r = \begin{bmatrix} r \cdot 3^r & 2^r \\ 0 & (r-1)3^r \end{bmatrix}$ be 3 given matrices
- Compute the value of $\sum_{r=1}^{50} \text{tr}((AB)^r C_r)$. (where $\text{tr}(A)$ denotes trace of matrix A)
13. Consider the two matrices A and B where $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$; $B = \begin{bmatrix} 5 \\ -3 \end{bmatrix}$. Let $n(A)$ denotes the number of elements in A and $n(XY) = 0$, when the two matrices X and Y are not conformable for multiplication. If $C = (AB)(B'A)$; $D = (B'A)(AB)$ then, find the value of $\left(\frac{n(C)((|D|^2 + n(D)))}{n(A) - n(B)} \right)$.
14. Given $A = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$; $B = \begin{bmatrix} 9 & 3 \\ 3 & 1 \end{bmatrix}$. I is a unit matrix of order 2. Find all possible matrix X in the following cases.
- (a) $AX = A$ (b) $XA = I$ (c) $XB = \mathbf{0}$ but $BX \neq \mathbf{0}$.
15. Find the product of two matrices A and B , where $A = \begin{bmatrix} -5 & 1 & 3 \\ 7 & 1 & -5 \\ 1 & -1 & 1 \end{bmatrix}$ & $B = \begin{bmatrix} 1 & 1 & 2 \\ 3 & 2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and use it to solve the following system of linear equations, $x + y + 2z = 1$; $3x + 2y + z = 7$; $2x + y + 3z = 2$.



ANSWER KEY

1. (a) 4, (b) 8
3. (i) 5, (ii) 18, (iii) 8, (iv) 5
4. 2
5. 64
6. 36
7. 0
8. $\begin{bmatrix} 17 & 4 & -19 \\ -10 & 0 & 13 \\ -21 & -3 & 25 \end{bmatrix}$
9. (a) $s \begin{bmatrix} -4 & 7 & -7 \\ 3 & -5 & 5 \end{bmatrix}$; (b) $\frac{1}{19} \begin{bmatrix} 48 & -25 \\ -70 & 42 \end{bmatrix}$
10. 1
11. $x_1 = 1, x_2 = -1, x_3 = 1$
12. $3(49.3^{50} + 1)$
13. 650
14. (i) $X = \begin{bmatrix} a & b \\ 2-2a & 1-2b \end{bmatrix}$ for $a, b \in \mathbb{R}$; (ii) X does not exist;
- (iii) $X = \begin{bmatrix} a & -3a \\ c & -3c \end{bmatrix}$ where $a, c \in \mathbb{R}$ and $3a + c \neq 0$; $3b + d \neq 0$
15. $x = 2, y = 1, z = -1$

1. Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{pmatrix}$. If u_1 and u_2 are column matrices such that $Au_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ and $Au_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, then $u_1 + u_2$ is equal to :

[AIEEE-2012]

- (1) $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ (2) $\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$ (3) $\begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ (4) $\begin{pmatrix} -1 \\ -1 \\ 0 \end{pmatrix}$

2. If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3×3 matrix A and $|A| = 4$, then α is equal to

[JEE(Main)-2013]

- (1) 4 (2) 11 (3) 5 (4) 0

3. If A is an 3×3 non-singular matrix such that $AA' = A'A$ and $B = A^{-1}A'$, the BB' equals:

[JEE(Main)-2014]

- (1) $I + B$ (2) I (3) B^{-1} (4) $(B^{-1})'$

4. If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to:

[JEE(Main)-2015]

- (1) (2, 1) (2) (-2, -1) (3) (2, -1) (4) (-2, 1)

5. If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $\text{Adj } A = AA^T$, then $5a + b$ is equal to:

[JEE(Main)-2016]

- (1) 13 (2) -1 (3) 5 (4) 4

6. If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then $\text{adj } (3A^2 + 12A)$ is equal to:

[JEE(Main)-2017]

- (1) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$ (2) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$
(3) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$ (4) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$



7. If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then the matrix A^{-50} when $\theta = \frac{\pi}{12}$, is equal to: **[JEE(Main)-2019]**

(1) $\begin{bmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$ (2) $\begin{bmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$ (3) $\begin{bmatrix} \frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & \frac{1}{2} \end{bmatrix}$ (4) $\begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}$

8. If $A = \begin{bmatrix} e^t & e^{-t} \cosh t & e^{-t} \sinh t \\ e^t & -e^{-t} \cosh t - e^{-t} \sinh t & -e^{-t} \sinh t + e^{-t} \cosh t \\ e^t & 2e^{-t} \sinh t & -2e^{-t} \cosh t \end{bmatrix}$ then A is: **[JEE(Main)-2019]**

- (1) invertible for all $t \in \mathbb{R}$ (2) invertible only if $t = \frac{\pi}{2}$
(3) not invertible for any $t \in \mathbb{R}$ (4) invertible only if $t = \pi$

9. Let $A = \begin{pmatrix} 0 & 2q & r \\ p & q & -r \\ p & -q & r \end{pmatrix}$. If $AA^T = I_3$, then $|p|$ is: **[JEE(Main)-2019]**

(1) $\frac{1}{\sqrt{2}}$ (2) $\frac{1}{\sqrt{6}}$ (3) $\frac{1}{\sqrt{3}}$ (4) $\frac{1}{\sqrt{5}}$

10. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 9 & 3 & 1 \end{bmatrix}$ and $Q = [q_{ij}]$ be two 3×3 matrices such that $Q - P^5 = I_3$. Then $\frac{q_{21} + q_{31}}{q_{32}}$ is equal to: **[JEE(Main)-2019]**

(1) 15 (2) 9 (3) 10 (4) 135

11. Let $A = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}$, ($\alpha \in \mathbb{R}$) such that $A^{32} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Then a value of α is: **[JEE(Main)-2019]**

(1) 0 (2) $\frac{\pi}{16}$ (3) $\frac{\pi}{64}$ (4) $\frac{\pi}{32}$

12. The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$, ($x, y \in \mathbb{R}, x \neq y$) for which $A^T A = 3I_3$ is: **[JEE(Main)-2019]**

(1) 3 (2) 2 (3) 4 (4) 6

13. If A is a symmetric matrix and B is a skew-symmetric matrix such that $A + B = \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix}$, then AB is equal to: **[JEE(Main)-2019]**

(1) $\begin{bmatrix} 4 & -2 \\ -1 & -4 \end{bmatrix}$ (2) $\begin{bmatrix} -4 & -2 \\ -1 & 4 \end{bmatrix}$ (3) $\begin{bmatrix} 4 & -2 \\ 1 & -4 \end{bmatrix}$ (4) $\begin{bmatrix} -4 & 2 \\ 1 & 4 \end{bmatrix}$



14. Let A and B be two invertible matrices of order 3×3 . If $\det(ABA^T) = 8$ and $\det(AB^{-1}) = 8$, then $\det(BA^{-1}B^T)$ is equal to :

[JEE(Main)-2019]

- (1) $\frac{1}{4}$ (2) $\frac{1}{16}$ (3) 1 (4) 16

15. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two 3×3 real matrices such that $b_{ij} = (3)^{(i+j-2)}a_{ji}$, where $i, j = 1, 2, 3$. If the determinant of B is 81, then the determinant of A is ;

[JEE(Main)-2020]

- (1) $1/9$ (2) $1/81$ (3) 3 (4) $1/3$

16. The number of all 3×3 matrices A, with entries from the set $\{-1, 0, 1\}$ such that the sum of the diagonal elements of AA^T is 3, is _____.

[JEE(Main)-2020]

17. If $A = \begin{pmatrix} 2 & 2 \\ 9 & 4 \end{pmatrix}$ and $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ then $10A^{-1}$ is equal to:

[JEE(Main)-2020]

- (1) $A - 6I$ (2) $4I - A$ (3) $A - 4I$ (4) $6I - A$

18. Let $A = \begin{bmatrix} x & 1 \\ 1 & 0 \end{bmatrix}$, $x \in \mathbb{R}$ and $A^4 = [a_{ij}]$. If $a_{11} = 109$, then a_{22} is equal to _____.

[JEE(Main)-2020]

19. Let A be a 3×3 matrix such that $\text{adj } A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & -2 & -1 \end{bmatrix}$ and $B = \text{adj}(\text{adj } A)$.

If $|A| = \lambda$ and $|(B^{-1})^T| = \mu$, then the ordered pair, $(|\lambda|, \mu)$ is equal to:

[JEE(Main)-2020]

- (1) $\left(9, \frac{1}{9}\right)$ (2) $(3, 81)$ (3) $\left(9, \frac{1}{81}\right)$ (4) $\left(3, \frac{1}{81}\right)$

20. Let $\lambda \in \mathbb{R}$. The system of linear equations

$$2x_1 - 4x_2 + \lambda x_3 = 1$$

$$x_1 - 6x_2 + x_3 = 2$$

$$\lambda x_1 - 10x_2 + 4x_3 = 3$$

is inconsistent for:

[JEE(Main)-2020]

- (1) exactly two values of λ . (2) exactly one positive value of λ .
(3) every value of λ . (4) exactly one negative value of λ .

21. Let $P = \begin{bmatrix} 2 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$. Suppose $Q = [q_{ij}]$ is a matrix satisfying $PQ = kI_3$ for some

non-zero $k \in \mathbb{R}$. If $q_{23} = -\frac{k}{8}$ and $|Q| = \frac{k^2}{2}$, then $\alpha^2 + k^2$ is equal to:

[JEE(Main)-2021]



- 22.** Let A and B be 3×3 real matrices such that A is symmetric matrix and B is skew-symmetric matrix. Then the system of linear equations $(A^2B^2 - B^2A^2)X = 0$, where X is a 3×1 column matrix of unknown variables and O is a 3×1 null matrix, has: **[JEE(Main)-2021]**

- (1) a unique solution (2) exactly solutions
(3) infinitely solutions (4) no solution

- 23.** If $A = \begin{bmatrix} 0 & -\tan\left(\frac{\theta}{2}\right) \\ \tan\left(\frac{\theta}{2}\right) & 0 \end{bmatrix}$ and $(I_2 + A)(I_2 - A)^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then $13(a^2 + b^2)$ is equal to:

[JEE(Main)-2021]

- 24.** Let $A = \begin{bmatrix} x & y & z \\ y & z & x \\ z & x & y \end{bmatrix}$, where x, y and z are real numbers such that $x + y + z > 0$ and $xyz = 2$. If

$A^2 = I_3$, then the value of $x^3 + y^3 + z^3$ is:

[JEE(Main)-2021]

- 25.** If for the matrix, $A = \begin{bmatrix} 1 & -\alpha \\ \alpha & \beta \end{bmatrix}$, $AA^T = I_2$, then the value of $\alpha^4 + \beta^4$ is:

[JEE(Main)-2021]

- (1) 1 (2) 3 (3) 2 (4) 4

- 26.** Let $A = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}$ and $B = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ be two 2×1 matrices with real entries such that $A = XB$, where $X = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & -1 \\ 1 & k \end{bmatrix}$, and $k \in \mathbb{R}$. If $a_1^2 + a_2^2 = \frac{2}{3}(b_1^2 + b_2^2)$ and $(k^2 + 1)b_2^2 \neq -2b_1b_2$, then the value of k is _____.

[JEE(Main)-2021]

- 27.** Let $S = \{\sqrt{n} : 1 \leq n \leq 50 \text{ and } n \text{ is odd}\}$ and $a \in S$ and $A = \begin{bmatrix} 1 & 0 & a \\ -1 & 1 & 0 \\ -a & 0 & 1 \end{bmatrix}$. If $\sum_{a \in S} \det(\text{adj } A) = 100\lambda$, then

λ is equal to:

[JEE(Main)-2022]

- (1) 218 (2) 221 (3) 663 (4) 1717

- 28.** Let $A = \begin{pmatrix} 2 & -2 \\ 1 & -1 \end{pmatrix}$ and $B = \begin{pmatrix} -1 & 2 \\ -1 & 2 \end{pmatrix}$. Then the number of elements in the set $\{(n, m) : n, m \in \{1, 2, \dots, 10\} \text{ and } nA^n + mB^m = I\}$ is _____.

[JEE(Main)-2022]



29. Let $X = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, $Y = \alpha I + \beta X + \gamma X^2$ and $Z = \alpha^2 I - \alpha\beta X + (\beta^2 - \alpha\gamma)X^2$, $\alpha, \beta, \gamma \in \mathbb{R}$. If $Y^{-1} = \begin{bmatrix} \frac{1}{5} & \frac{-2}{5} & \frac{1}{5} \\ 0 & \frac{1}{5} & \frac{-2}{5} \\ 0 & 0 & \frac{1}{5} \end{bmatrix}$,

then $(\alpha - \beta + \gamma)$ is equal to _____.

[JEE(Main)-2022]

30. The positive value of the determinant of the matrix A, whose $\text{Adj}(\text{Adj}(A)) = \begin{pmatrix} 14 & 28 & -14 \\ -14 & 14 & 28 \\ 28 & -14 & 14 \end{pmatrix}$, is:

[JEE(Main)-2022]

31. Let A be a matrix of order 2×2 , whose entries are from the set $\{0, 1, 2, 3, 4, 5\}$. If the sum of all the entries of A is a prime number p, $2 < p < 8$, then the number of such matrices A is:

[JEE(Main)-2022]

32. Let $A = [a_{ij}]$ be a square matrix of order 3 such that $a_{ij} = 2^{j-i}$, for all $i, j = 1, 2, 3$. Then, the matrix $A^2 + A^3 + \dots + A^{10}$ is equal to:

[JEE(Main)-2022]

(1) $\left(\frac{3^{10}-3}{2}\right)A$ (2) $\left(\frac{3^{10}-1}{2}\right)A$ (3) $\left(\frac{3^{10}+1}{2}\right)A$ (4) $\left(\frac{3^{10}+3}{2}\right)A$

ANSWER KEY

1.	(1)	2.	(2)	3.	(2)	4.	(2)	5.	(3)	6.	(3)	7.	(1)
8.	(1)	9.	(1)	10.	(3)	11.	(3)	12.	(3)	13.	(1)	14.	(2)
15.	(1)	16.	672	17.	(1)	18.	10.00	19.	(3)	20.	(4)	21.	7
22.	(3)	23.	13	24.	7	25.	(1)	26.	1.00	27.	(B)	28.	1
29.	100	30.	14	31.	180	32.	(A)						

JEE-Advanced (Previous Year Questions)

1. Let $P = [a_{ij}]$ be a 3×3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j}a_{ij}$ for $1 \leq i, j \leq 3$. If the determinant of P is 2, then the determinant of the matrix Q is – **[JEE(Advanced)-2012]**
 (A) 2^{10} (B) 2^{11} (C) 2^{12} (D) 2^{13}

2. If P is a 3×3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3×3 identity matrix, then there exists a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that **[JEE(Advanced)-2012]**
 (A) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (B) $PX = X$ (C) $PX = 2X$ (D) $PX = -X$

3. If the adjoint of a 3×3 matrix P is $\begin{bmatrix} 1 & 4 & 4 \\ 2 & 1 & 7 \\ 1 & 1 & 3 \end{bmatrix}$, then the possible value(s) of the determinant of P is (are) – **[JEE(Advanced)-2012]**
 (A) -2 (B) -1 (C) 1 (D) 2

4. For 3×3 matrices M and N , which of the following statement(s) is (are) NOT correct?
 (A) $N^T M N$ is symmetric or skew symmetric, according as M is symmetric or skew symmetric
 (B) $MN - NM$ is skew symmetric for all symmetric matrices M and N
 (C) MN is symmetric for all symmetric matrices M and N
 (D) $(\text{adj } M)(\text{adj } N) = \text{adj } (MN)$ for all invertible matrices M and N **[JEE(Advanced)-2013]**

5. Let M be a 2×2 symmetric matrix with integer entries. Then M is invertible if
 (A) the first column of M is the transpose of the second row of M
 (B) the second row of M is the transpose of the first column of M
 (C) M is a diagonal matrix with nonzero entries in the main diagonal
 (D) the product of entries in the main diagonal of M is not the square of an integer **[JEE(Advanced)-2014]**

6. Let M and N be two 3×3 matrices such that $MN = NM$. Further, if $M \neq N^2$ and $M^2 = N^4$, then
 (A) determinant of $(M^2 + MN^2)$ is 0 **[JEE(Advanced)-2014]**
 (B) there is a 3×3 non-zero matrix U such that $(M^2 + MN^2)U$ is zero matrix
 (C) determinant of $(M^2 + MN^2) \geq 1$
 (D) for a 3×3 matrix U , if $(M^2 + MN^2)U$ equals the zero matrix then U is the zero matrix



7. Let X and Y be two arbitrary, 3×3 , non-zero, skew-symmetric matrices and Z be an arbitrary 3×3 , non-zero, symmetric matrix. Then which of the following matrices is (are) skew symmetric?

[JEE(Advanced)-2015]

- (A) $Y^3Z^4 - Z^4Y^3$ (B) $X^{44} + Y^{44}$ (C) $X^4Z^3 - Z^3X^4$ (D) $X^{23} + Y^{23}$

8. Let $P = \begin{bmatrix} 3 & -1 & -2 \\ 2 & 0 & \alpha \\ 3 & -5 & 0 \end{bmatrix}$, where $\alpha \in \mathbb{R}$, Suppose $Q = [q_{ij}]$ is a matrix such that $PQ = kI$, where

$k \in \mathbb{R}$, $k \neq 0$ and I is the identity matrix of order 3. If $q_{23} = -\frac{k}{8}$ and $\det(Q) = \frac{k^2}{2}$, then-

[JEE(Advanced)-2016]

- (A) $\alpha = 0$, $k = 8$ (B) $4\alpha - k + 8 = 0$ (C) $\det(\text{Padj}(Q)) = 2^9$ (D) $\det(\text{Qadj}(P)) = 2^{13}$

9. Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix such that

$P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals

[JEE(Advanced)-2016]

- (A) 52 (B) 103 (C) 201 (D) 205

10. Which of the following is (are) NOT the square of a 3×3 matrix with real entries?

[JEE(Advanced)-2017]

- (A) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (B) $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ (C) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ (D) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

11. How many 3×3 matrices M with entries from $\{0, 1, 2\}$ are there, for which the sum of the diagonal entries of $M^T M$ is 5?

[JEE(Advanced)-2017]

- (A) 198 (B) 135 (C) 126 (D) 162

12. For a real number α , if the system $\begin{bmatrix} 1 & \alpha & \alpha^2 \\ \alpha & 1 & \alpha \\ \alpha^2 & \alpha & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

of linear equations, has infinitely many solutions, then $1 + \alpha + \alpha^2 =$ [JEE(Advanced)-2017]



13. Let S be the set of all column matrices $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ such that $b_1, b_2, b_3 \in \mathbb{R}$ and the system of

equations (in real variables)

$$-x + 2y + 5z = b_1$$

$$2x - 4y + 3z = b_2$$

$$x - 2y + 2z = b_3$$

has at least one solution. Then, which of the following system(s) (in real variables) has(have) at

least one solution for each $\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} \in S$?

[JEE(Advanced)-2018]

(A) $x + 2y + 3z = b_1$, $4y + 5z = b_2$ and $x + 2y + 6z = b_3$

(B) $x + y + 3z = b_1$, $5x + 2y + 6z = b_2$ and $-2x - y - 3z = b_3$

(C) $-x + 2y - 5z = b_1$, $2x - 4y + 10z = b_2$ and $x - 2y + 5z = b_3$

(D) $x + 2y + 5z = b_1$, $2x + 3z = b_2$ and $x + 4y - 5z = b_3$

14. Let P be a matrix of order 3×3 such that all the entries in P are from the set $\{-1, 0, 1\}$. Then, the maximum possible value of the determinant of P is ____.

[JEE(Advanced)-2018]

15. Let $M = \begin{bmatrix} \sin^4 \theta & -1 - \sin^2 \theta \\ 1 + \cos^2 \theta & \cos^4 \theta \end{bmatrix} = \alpha I + \beta M^{-1}$,

where $\alpha = \alpha(\theta)$ and $\beta = \beta(\theta)$ are real numbers, and I is the 2×2 identity matrix. If

α is the minimum of the set $\{\alpha(\theta) : \theta \in [0, 2\pi)\}$ and

β is the minimum of the set $\{\beta(\theta) : \theta \in [0, 2\pi)\}$,

then the value of $\alpha + \beta$ is

[JEE(Advanced)-2019]

(A) $-\frac{31}{16}$

(B) $-\frac{29}{16}$

(C) $-\frac{17}{16}$

(D) $-\frac{37}{16}$

16. Let $M = \begin{bmatrix} 0 & 1 & a \\ 1 & 2 & 3 \\ 3 & b & 1 \end{bmatrix}$ and $\text{adj } M = \begin{bmatrix} -1 & 1 & -1 \\ 8 & -6 & 2 \\ -5 & 3 & -1 \end{bmatrix}$ where a and b are real numbers. Which of the following options is/are correct?

[JEE(Advanced)-2019]

(A) $a + b = 3$

(B) If $M \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, then $\alpha - \beta + \gamma = 3$

(C) $(\text{adj } M)^{-1} + \text{adj } M^{-1} = -M$

(D) $\det(\text{adj } M^2) = 81$



17. Let $x \in \mathbb{R}$ and let

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \quad Q = \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 6 \end{bmatrix} \text{ and } R = PQP^{-1}.$$

Then which of the following options is/are correct?

[JEE(Advanced)-2019]

(A) For $x = 0$, if $R \begin{bmatrix} 1 \\ a \\ b \end{bmatrix} = 6 \begin{bmatrix} 1 \\ a \\ b \end{bmatrix}$, then $a + b = 5$

(B) $\det R = \det \begin{bmatrix} 2 & x & x \\ 0 & 4 & 0 \\ x & x & 5 \end{bmatrix} + 8$, for all $x \in \mathbb{R}$

(C) There exists a real number x such that $PQ = QP$

(D) For $x = 1$, there exists a unit vector $\alpha \hat{i} + \beta \hat{j} + \gamma \hat{k}$ for which $R \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

18. Let M be a 3×3 invertible matrix with real entries and let I denote the 3×3 identity matrix. If $M^{-1} = \text{adj}(\text{adj } M)$, then which of the following statement is/are ALWAYS TRUE?

(A) $M = I$

(B) $\det M = 1$

(C) $M^2 = I$

(D) $(\text{adj } M)^2 = I$

[JEE(Advanced)-2020]

19. The trace of a square matrix is defined to be the sum of its diagonal entries. If A is a 2×2 matrix such that the trace of A is 3 and the trace of A^3 is -18 , then the value of the determinant of A is _____

[JEE(Advanced)-2020]

Question Stem for Question Nos. 20 and 21

Question Stem

Let α, β and γ be real numbers such that the system of linear equations

$$x + 2y + 3z = \alpha$$

$$4x + 5y + 6z = \beta$$

$$7x + 8y + 9z = \gamma - 1$$

is consistent. Let $|M|$ represent the determinant of the matrix

[JEE(Advanced)-2021]

$$M = \begin{bmatrix} \alpha & 2 & \gamma \\ \beta & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Let P be the plane containing all those (α, β, γ) for which the above system of linear equation is consistent, and D be the square of the distance of the point $(0, 1, 0)$ from the plane P .

20. The value of $|M|$ is _____.

21. The value of D is _____.



22. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let

$$E = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 8 & 13 & 18 \end{bmatrix}, P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \text{ and } F = \begin{bmatrix} 1 & 3 & 2 \\ 8 & 18 & 13 \\ 2 & 4 & 3 \end{bmatrix}$$

If Q is a nonsingular matrix of order 3×3 , then which of the following statements is (are) TRUE?

[JEE(Advanced)-2021]

(A) $F = PEP$ and $P^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(B) $|EQ + PFQ^{-1}| = |EQ| + |PFQ^{-1}|$

(C) $|(EF)^3| > |EF|^2$

(D) Sum of the diagonal entries of $P^{-1}EP + F$ is equal to the sum of diagonal entries of $E + P^{-1}FP$

23. For any 3×3 matrix M , let $|M|$ denote the determinant of M . Let I be the 3×3 identity matrix. Let E and F be two 3×3 matrices such that $(I - EF)$ is invertible. If $G = (I - EF)^{-1}$, then which of the following statements is (are) TRUE?

[JEE(Advanced)-2021]

(A) $|FE| = |I - FE||FGE|$

(B) $(I - FE)(I + FGE) = I$

(C) $EFG = GEF$

(D) $(I - FE)(I - FGE) = I$

24. Let β be a real number. Consider the matrix

[JEE(Advanced)-2022]

$$A = \begin{pmatrix} \beta & 0 & 1 \\ 2 & 1 & -2 \\ 3 & 1 & -2 \end{pmatrix}.$$

If $A^7 - (\beta - 1)A^6 - \beta A^5$ is a singular matrix, then the value of 9β is _____.

25. If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ \frac{2}{3} & \frac{2}{1} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2022} ?

[JEE(Advanced)-2022]

(A) $\begin{pmatrix} 3034 & 3033 \\ -3033 & -3032 \end{pmatrix}$

(B) $\begin{pmatrix} 3034 & -3033 \\ -3033 & -3032 \end{pmatrix}$

(C) $\begin{pmatrix} 3033 & 3032 \\ -3032 & -3031 \end{pmatrix}$

(D) $\begin{pmatrix} 3032 & 3031 \\ -3031 & -3030 \end{pmatrix}$

ANSWER KEY

- | | | | | | | | | | | | | | |
|-----|-------|-----|-------|-----|------|-----|-------|-----|------|-----|------|-----|------|
| 1. | (D) | 2. | (D) | 3. | (AD) | 4. | (CD) | 5. | (CD) | 6. | (AB) | 7. | (CD) |
| 8. | (BC) | 9. | (B) | 10. | (AB) | 11. | (A) | 12. | 1 | 13. | (AD) | 14. | 4.00 |
| 15. | (B) | 16. | (ABC) | 17. | (AB) | 18. | (BCD) | 19. | 5 | 20. | 1 | 21. | 1.5 |
| 22. | (ABD) | 23. | (ABC) | 24. | 3 | 25. | (A) | | | | | | |

